

Review

1.

$$2 \int_0^2 (x^3 - 3x^2 - 2x) dx$$

$$= \left[\frac{1}{4}x^4 - x^3 - x^2 \right]_0^2$$

$$= \left(\frac{16}{4} - 8 - 4 \right) - (0)$$

$$= -8$$

$$3 \quad y = x^3 + px^2 + qx - 4$$

$$y' = 3x^2 + 2px + q$$

$$\text{At } x=2:$$

$$12 + 4p + q = 0$$

$$4p + q = -12$$

$$q = -12 - 4p$$

$$y' = 6x + 2p$$

$$\text{At } x=2:$$

$$12 + 2p = 0$$

$$p = -6$$

$$q = -12 - 4(-6)$$

$$= 12$$

$$4 \quad \log q = 0.3p - 1$$

$$q = 10^{0.3p - 1}$$

(c)

$$5 \quad \arccos\left(-\frac{1}{2}\right) = \pi - \arccos\left(\frac{1}{2}\right) \\ = \pi - \frac{\pi}{3} \\ = \frac{2\pi}{3}$$

$$\arcsin(-1) = -\frac{\pi}{2}$$

$$\therefore \frac{2\pi}{3} - \frac{\pi}{4} = \frac{5\pi}{12}$$

Consolidation

$$1 i) \int x e^{2x} dx = \frac{1}{2} x e^{2x} - \int \frac{1}{2} e^{2x} dx$$

$$= \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + c$$

$$ii) \int (1-2x) \sin x dx = (1-2x)(-\cos x) - \int 2 \cos x dx \\ = 2x \cos x - \cos x - 2 \sin x + c$$

$$2 i) \int_0^1 2x e^{-3x} dx = \left[\frac{2}{3} x e^{-3x} - \int \frac{2}{3} e^{-3x} dx \right]_0^1 \\ = \left[-\frac{2}{3} x e^{-3x} - \frac{2}{9} e^{-3x} \right]_0^1 \\ = -\frac{2}{3} e^{-3} - \frac{2}{9} e^{-3} - \left(0 - \frac{2}{9} \right) \\ = \frac{2}{9} - \frac{8}{9} e^{-3}$$

$$3 \int_0^3 \frac{x}{\sqrt{1+x}} dx \\ = \left[x(2(1+x)^{\frac{1}{2}}) - \int 2(1+x)^{\frac{1}{2}} dx \right]_0^3 \\ = \left[2x(1+x)^{\frac{1}{2}} - \frac{4}{3}(1+x)^{\frac{3}{2}} \right]_0^3 \\ = 6 \times 2 - \frac{32}{3} - \left(0 - \frac{4}{3} \right) \\ = 12 - \frac{32}{3} + \frac{4}{3} \\ = 12 - \frac{28}{3} \\ = \frac{8}{3}$$

$$4 i) \int_1^2 x^3 \ln 2x dx \\ = \left[\frac{1}{4} x^4 \ln 2x - \int \frac{1}{4} x^3 \times \frac{1}{x} dx \right]_1^2 \\ = \left[\frac{1}{4} x^4 \ln 2x - \frac{1}{16} x^4 \right]_1^2 \\ = \left(4 \ln 4 - \frac{1}{2} \right) - \left(\frac{1}{4} \ln 2 - \frac{1}{16} \right) \\ = 8 \ln 2 - \frac{1}{4} \ln 2 - \frac{1}{2} + \frac{1}{16} \\ = \frac{31}{4} \ln 2 - \frac{15}{16}$$

$$ii) \int_1^2 \ln x \, dx$$

$$= \int_1^2 1 \times \ln x \, dx$$

$$= \left[x \ln x \right]_1^2 + \int_1^2 1 \, dx$$

$$= \left[x \ln x + x \right]_1^2$$

$$= 2 \ln 2 + 2 - 1$$

$$= 2 \ln 2 + 1$$

$$6 \int_0^{\frac{\pi}{2}} x \sin 2x \, dx$$

$$= \left[-\frac{1}{2} x \cos 2x - \int -\frac{1}{2} \cos 2x \, dx \right]$$

$$\left[-\frac{1}{2} x \cos 2x + \frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= -\frac{\pi}{4} \times -1$$

$$= \frac{\pi}{4}$$

$$7 \int x^2 e^{(x^3)} \, dx$$

$$= \int \frac{x^3 e^u}{3x^2} \, du \quad u = x^3$$

$$= \int \frac{e^u}{3} \, du$$

$$= \frac{1}{3} e^u$$

$$= \frac{1}{3} e^{x^3} + C$$

$$\int x^2 e^{3x} \, dx$$

$$\frac{1}{3} x^2 e^{3x} - \int \frac{2}{3} x e^{3x} \, dx$$

$$= \frac{1}{3} x^2 e^{3x} - \left(\frac{1}{9} e^{3x} \times 2x - \int \frac{2}{9} e^{3x} \, dx \right)$$

$$= \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + C$$

Exam Q3

$$1) \int (2x+1) \ln x \, dx$$

$$= (x^2+x) \ln x - \int \frac{x^2+x}{x} \, dx$$

$$= x^2 \ln x + x \ln x - \frac{1}{2} x^2 - x + C$$

$$2) f(x) = (1-x)e^{2x} = e^{2x} - xe^{2x} \quad \begin{array}{l} e^{2x} = xe^{2x} \\ x=1 \end{array}$$

$$\text{At } x=0 \quad y=1$$

$$\text{At } y=0 \quad x=1$$

$$(i) f'(x) = e^{2x} - 2xe^{2x}$$

$$e^{2x} = 2xe^{2x} \quad \text{when } f'(x)=0$$

$$x = \frac{1}{2}$$

$$y = \frac{1}{2}e$$